Midterm Review

Computer Organization

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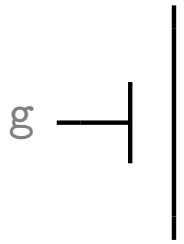
For each topic, we will **summarize** what you need to know, and then answer a few questions.

1. transistors
2. Boolean algebra
3. combinational design
4. logical devices
5. binary arithmetic
6. arithmetic circuits
7. flip-flops
8. state machines



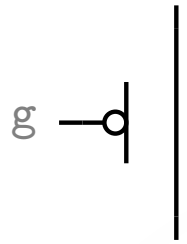
transistors

nMOS



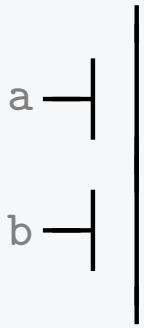
conducts when the gate is 1

pMOS



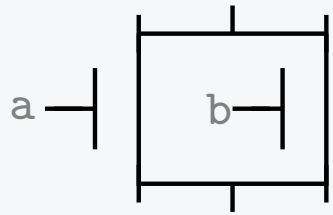
conducts when the gate is 0

in series



AND: both conduct

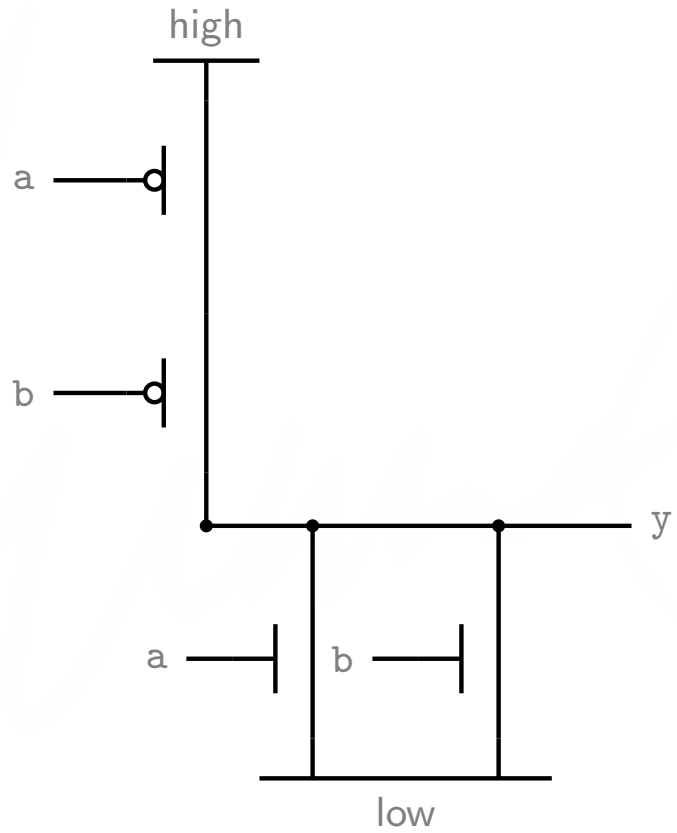
in parallel



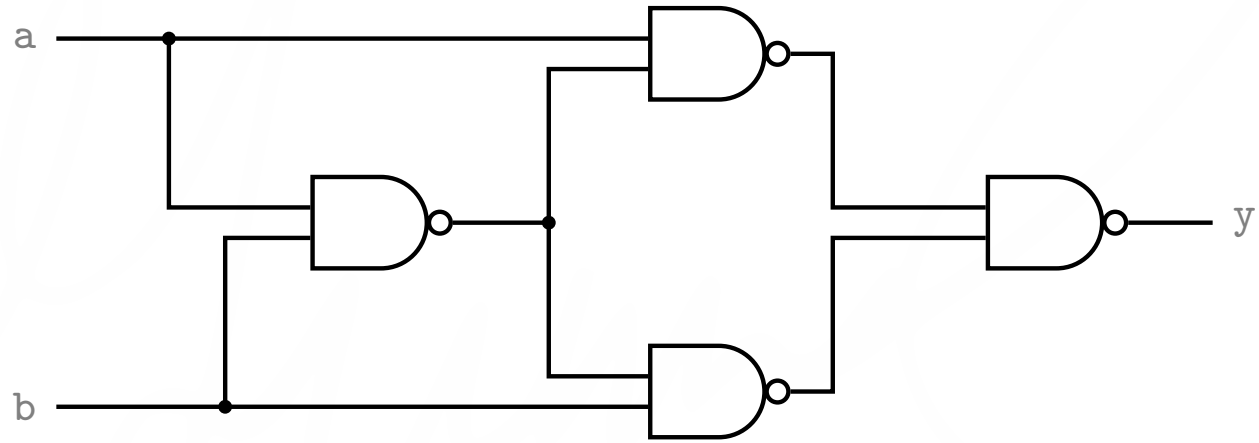
OR: either conducts

a gate connects its output to high through one network and to low through the other

question 1



question 2



2

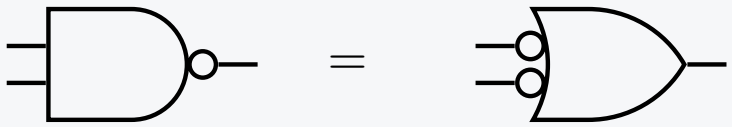
Boolean algebra

Each law comes in an AND form and an OR form.

identity	$a \cdot 1 = a$	$a + 0 = a$
null	$a \cdot 0 = 0$	$a + 1 = 1$
idempotent	$a \cdot a = a$	$a + a = a$
complement	$a \cdot \overline{a} = 0$	$a + \overline{a} = 1$
involution	$\overline{\overline{a}} = a$	
commutative	$ab = ba$	$a + b = b + a$
associative	$a(bc) = (ab)c$	$a + (b + c) = (a + b) + c$
distributive	$a(b + c) = ab + ac$	$a + bc = (a + b)(a + c)$
absorption	$a(a + b) = a$	$a + ab = a$
consensus	$ab + \overline{a}c + bc = ab + \overline{a}c$	
De Morgan	$\overline{a \cdot b} = \overline{a} + \overline{b}$	$\overline{a + b} = \overline{a} \cdot \overline{b}$

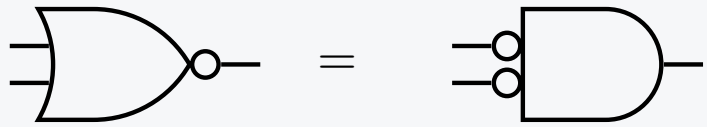
De Morgan's laws let us push a bubble through a gate: `AND` and `OR` swap, and every input and output flips.

first form



$$\overline{a \cdot b} = \overline{a} + \overline{b}$$

second form

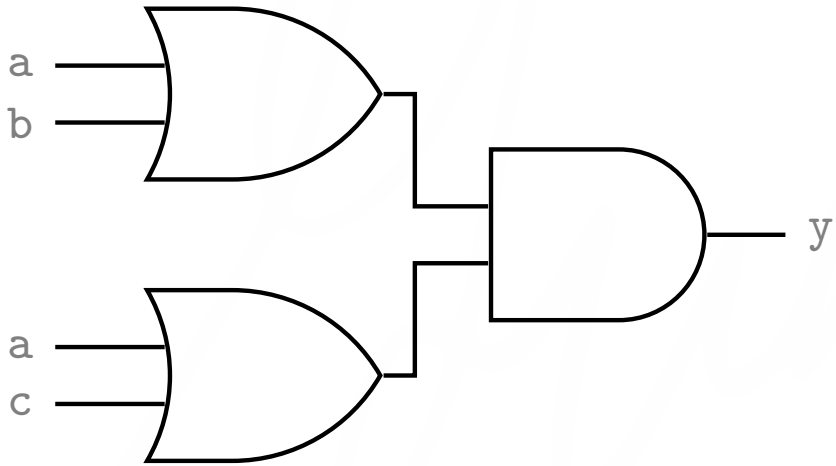


$$\overline{a + b} = \overline{a} \cdot \overline{b}$$

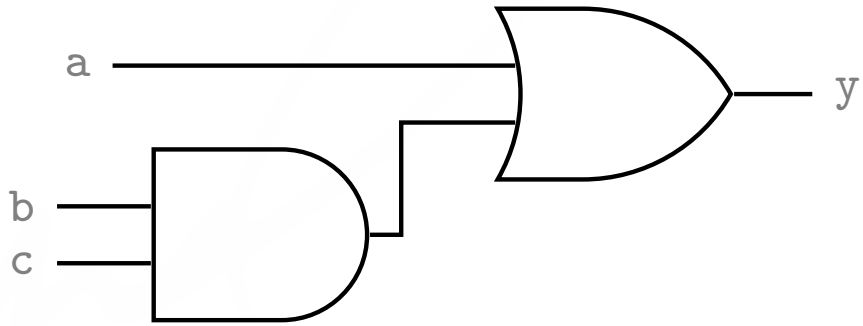
a `NAND` is an `OR` of inverted inputs; a `NOR` is an `AND` of inverted inputs

question 3

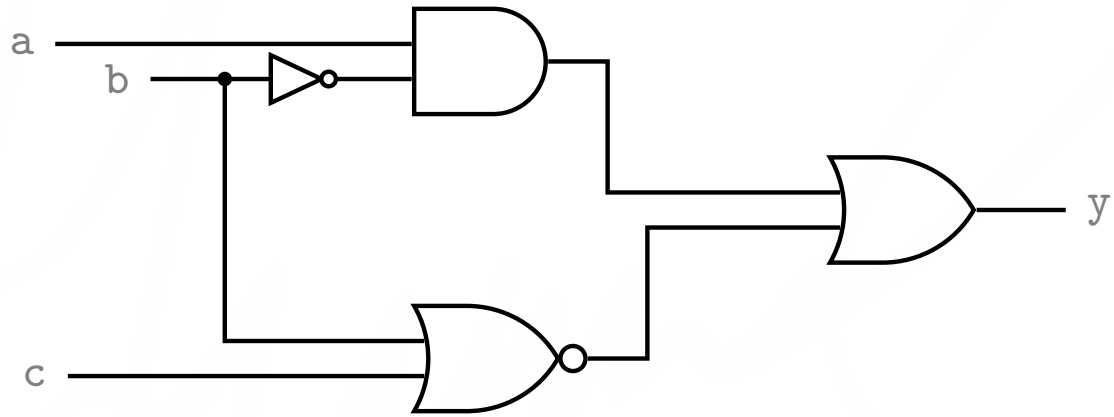
circuit A



circuit B

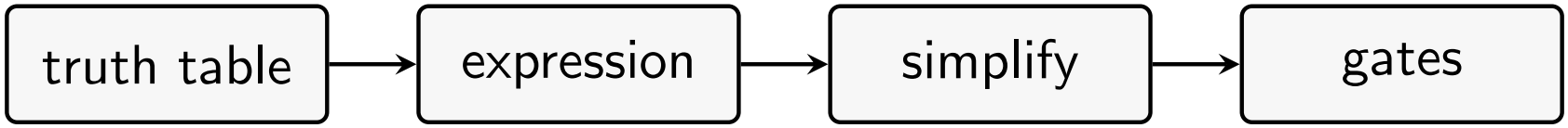


question 4



3

combinational design



a	b	c	f	
0	0	0	1	m_0
0	0	1	1	m_1
0	1	0	0	M_2
0	1	1	0	M_3
1	0	0	1	m_4
1	0	1	1	m_5
1	1	0	0	M_6
1	1	1	1	m_7

minterm m_i (the shaded rows)

AND of every input: $m_5 = \mathtt{a}\,\overline{\mathtt{b}}\,\mathtt{c}$

want these rows

$$\mathtt{f} = \Sigma m(0, 1, 4, 5, 7)$$

maxterm M_i (the white rows)

OR of every input: $M_2 = \mathtt{a} + \overline{\mathtt{b}} + \mathtt{c}$

don't want these rows

$$\mathtt{f} = \Pi M(2, 3, 6)$$

$M_i = \overline{m_i}$, so the two lists of the same function name opposite rows

sum of products

group the 1s

a \ bc	00	01	11	10
0	1	1	0	0
1	1	1	1	0

$$f = \bar{b} + ac$$

product of sums

group the 0s

a \ bc	00	01	11	10
0	1	1	0	0
1	1	1	1	0

$$f = (a + \bar{b})(\bar{b} + c)$$

size $1, 2, 4, 8, \ldots$

1	1	0	0
1	1	0	0

as large and as few as possible

wrap around

1	0	0	1
1	0	0	1

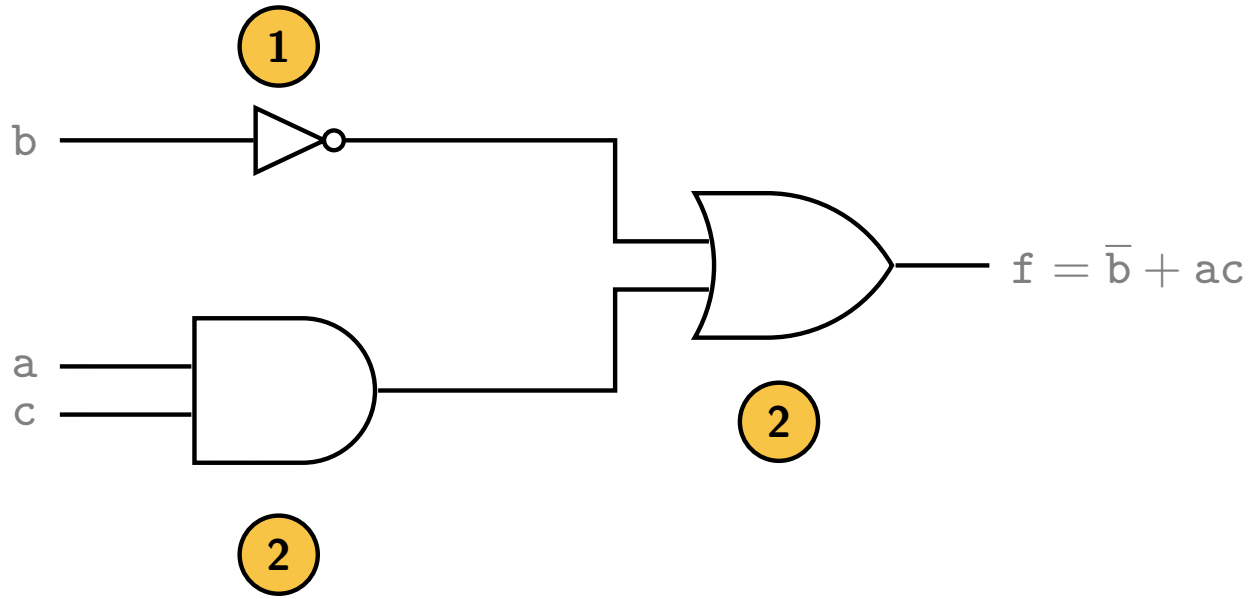
the edges are neighbours

don't-cares

0	1	X	0
0	1	1	0

an X joins a group if it helps

The **cost** of a circuit can be counted in three ways.



the circles count the inputs of each gate

L = 3

literals in the expression

$\bar{b}$, a, c

G = 4

gate inputs, not counting NOTs

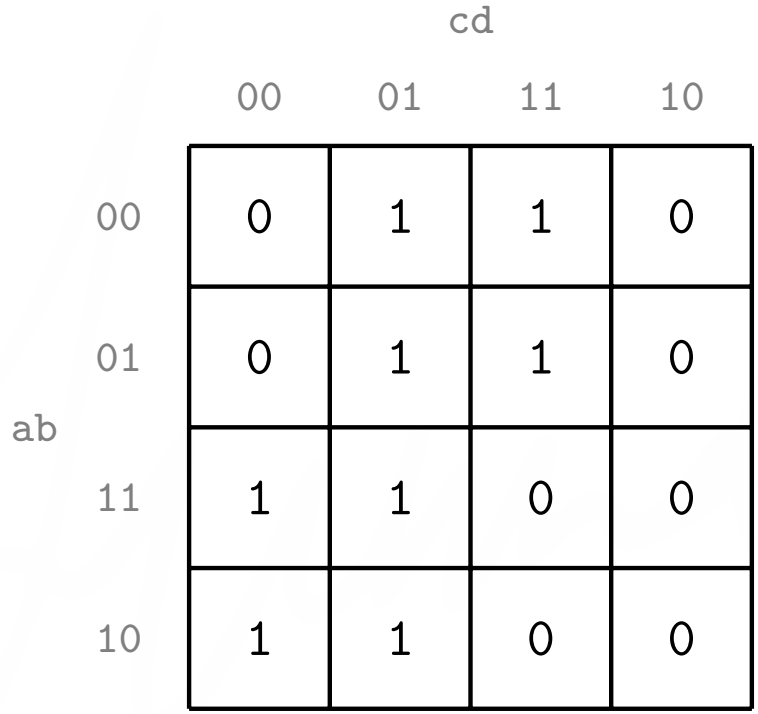
$2 + 2$

GN = 5

gate inputs, NOTs included

$2 + 2 + 1$

question 6



find the minimal sum of products

question 7

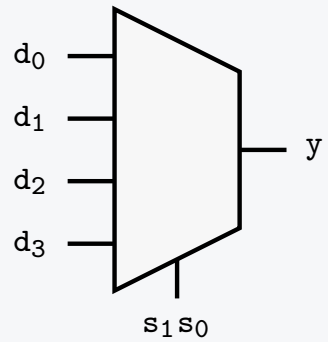
ab \ cd	00	01	11	10
00	1	0	0	1
01	0	0	0	0
11	0	0	0	0
10	1	1	X	X

find the minimal sum of products

4

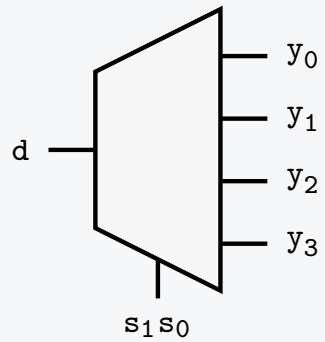
logical devices

multiplexer



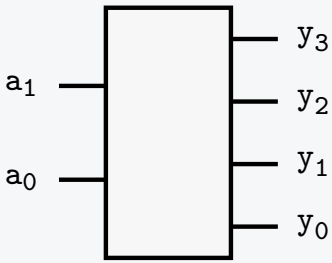
k select bits pick one of 2^k inputs

demultiplexer



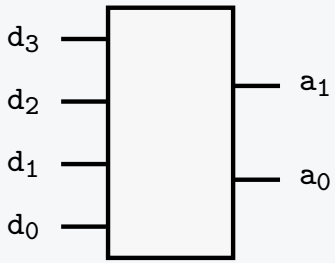
routes d to the selected output

decoder



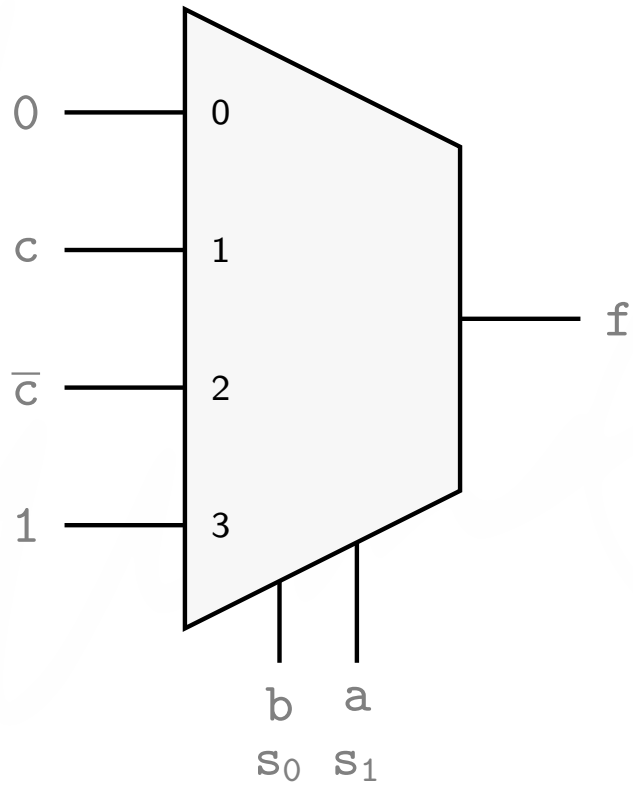
n bits $\rightarrow$ one of 2^n outputs is 1; $y_i = m_i$

encoder



the number of the one input that is 1

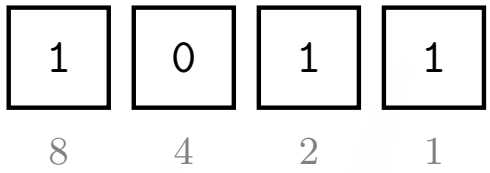
question 9



5

binary arithmetic

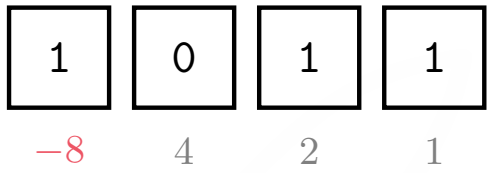
unsigned



$$8 + 2 + 1 = 11$$

range: 0 to $2^n - 1$

signed



$$-8 + 2 + 1 = -5$$

range: -2^{n-1} to $2^{n-1} - 1$

decimal to binary

$11 = 1011$

halve repeatedly; the remainders, read upward, are the bits

binary and hex

1011 0110 $\leftrightarrow$ B6

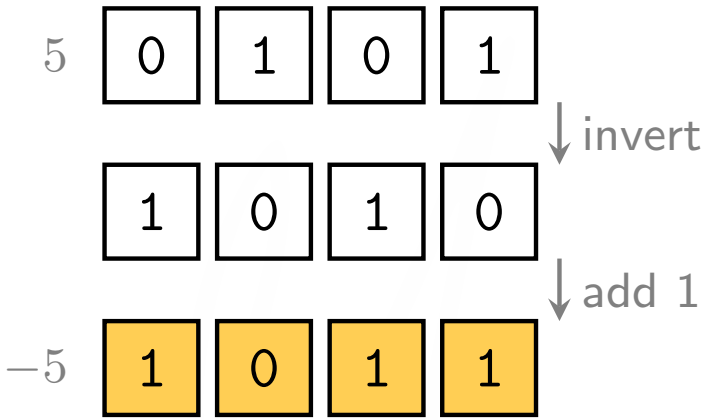
one hex symbol per group of 4 bits; A $= 10, \ldots,$ F $= 15$

binary to decimal

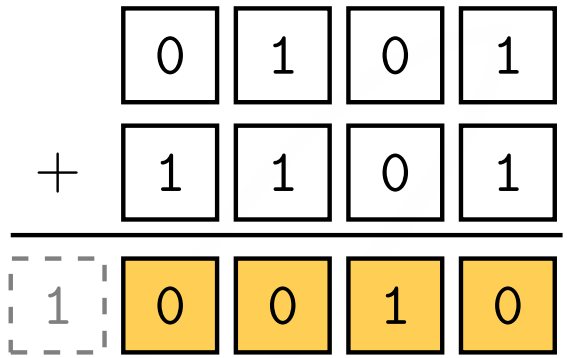
0110 $= 4 + 2 = 6$

add the weights of the 1s; signed, the top weight is negative

negating



adding



$5 + (-3) = 2$; the carry out is dropped

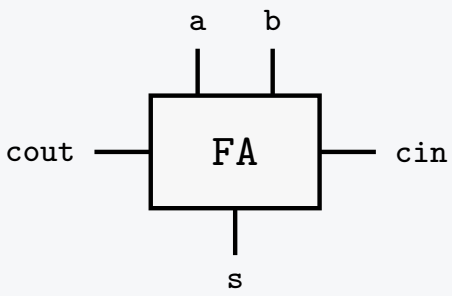
subtracting is adding the negation

$a - b = a + \overline{b} + 1$, the same bit operations for signed and unsigned numbers



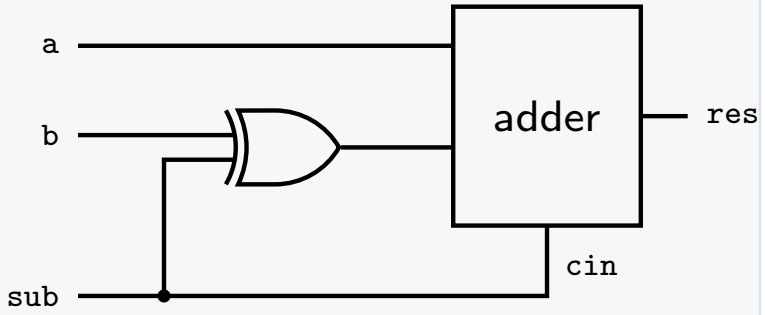
arithmetic circuits

full adder



$$\texttt{s} = \texttt{a} \oplus \texttt{b} \oplus \texttt{cin}$$
$$\texttt{cout} = \texttt{ab} + \texttt{cin}\,(\texttt{a} \oplus \texttt{b})$$

adder/subtractor



$$\texttt{sub} = 1\text{: } \texttt{a} + \overline{\texttt{b}} + 1 = \texttt{a} - \texttt{b}$$

unsigned: read `carry`

$$\begin{array}{r} 1100 \\ +\ 0110 \\ \hline 1\ 0010 \end{array}$$

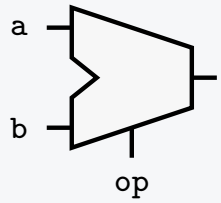
$12 + 6 \neq 2$: a + b is wrong when carry $= 1$
(a − b is wrong when carry $= 0$)

signed: read `overflow`

$$\begin{array}{r} 0110 \\ +\ 0011 \\ \hline 1001 \end{array}$$

$6 + 3 \neq -7$: two numbers of the same sign gave a result of the other sign

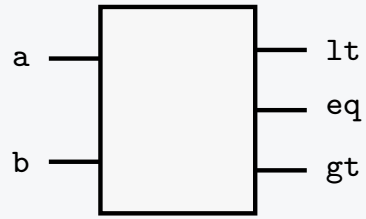
ALU



	useb	inv	cin
a	0	0	0
a + 1	0	0	1
a − 1	0	1	0
a + b	1	0	0
a − b	1	1	1

plus bitwise AND, OR, NOT;
a mux picks the result

comparator



1011 vs 1101: lt

the first bit from the left where they differ decides

shifter

1101 << 1 1010 $\times 2$

1101 >> 1 0110 logical

1101 >> 1 1110 arithmetic

multiplier

$$\begin{array}{r} 101 \\ \times\ 011 \\ \hline 101 \\ 1010 \\ \hline 1111 \end{array}$$

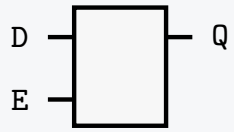
one shifted row per bit of b; the product needs $2n$ bits

signed: sign-extend, subtract the last row

7

latches and flip-flops

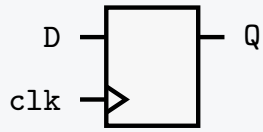
D latch



E	Q_t
0	Q_{t-1}
1	D

follows D while $E = 1$

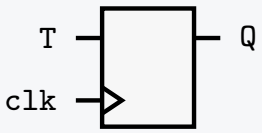
D flip-flop



D	Q_t
0	0
1	1

copies D on the clock edge

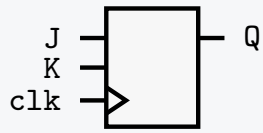
T flip-flop



T	Q_t
0	Q_{t-1}
1	$\overline{Q}_{t-1}$

toggles when $T = 1$: $D = T \oplus Q$

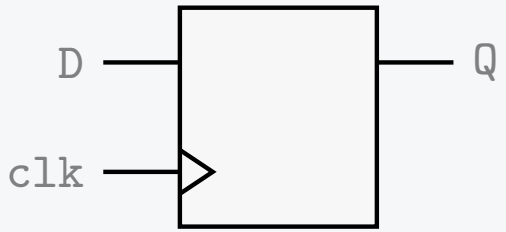
JK flip-flop



J	K	Q_t	
0	0	Q_{t-1}	hold
0	1	0	reset
1	0	1	set
1	1	$\overline{Q}_{t-1}$	toggle

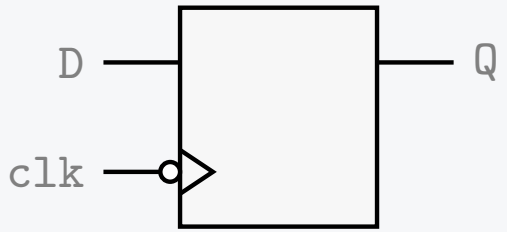
$D = J\overline{Q} + \overline{K}Q$

positive edge



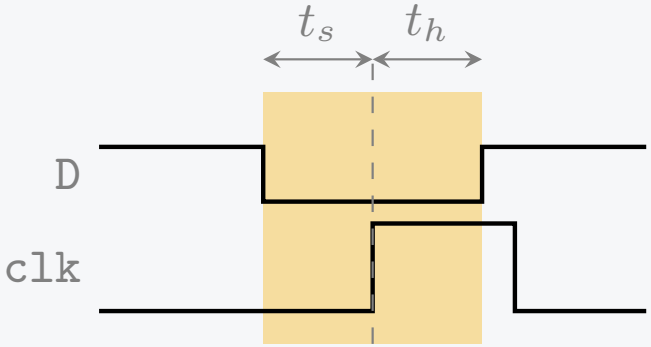
acts as clk rises

negative edge



acts as clk falls (the bubble)

setup and hold times

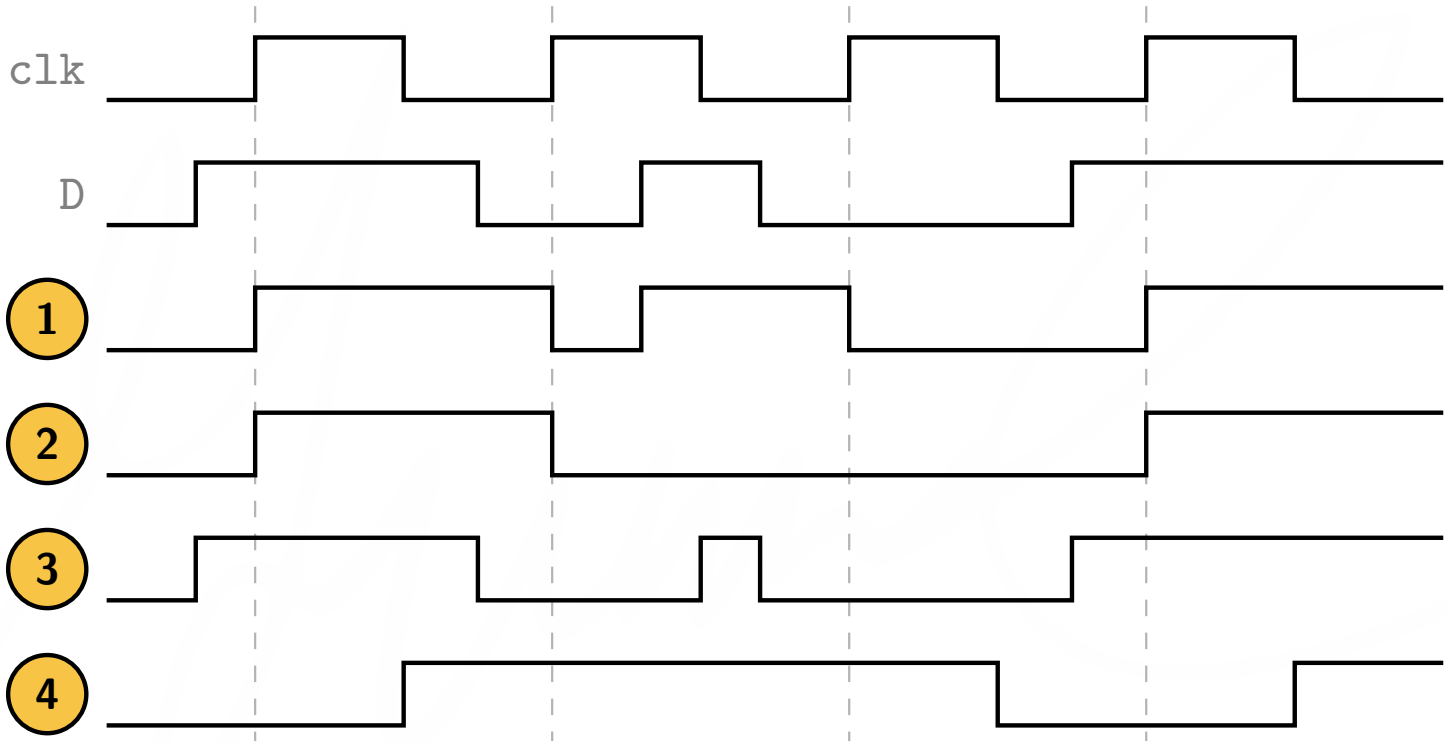


setup time t_s: D is stable this long *before* the edge

hold time t_h: D stays stable this long *after* the edge

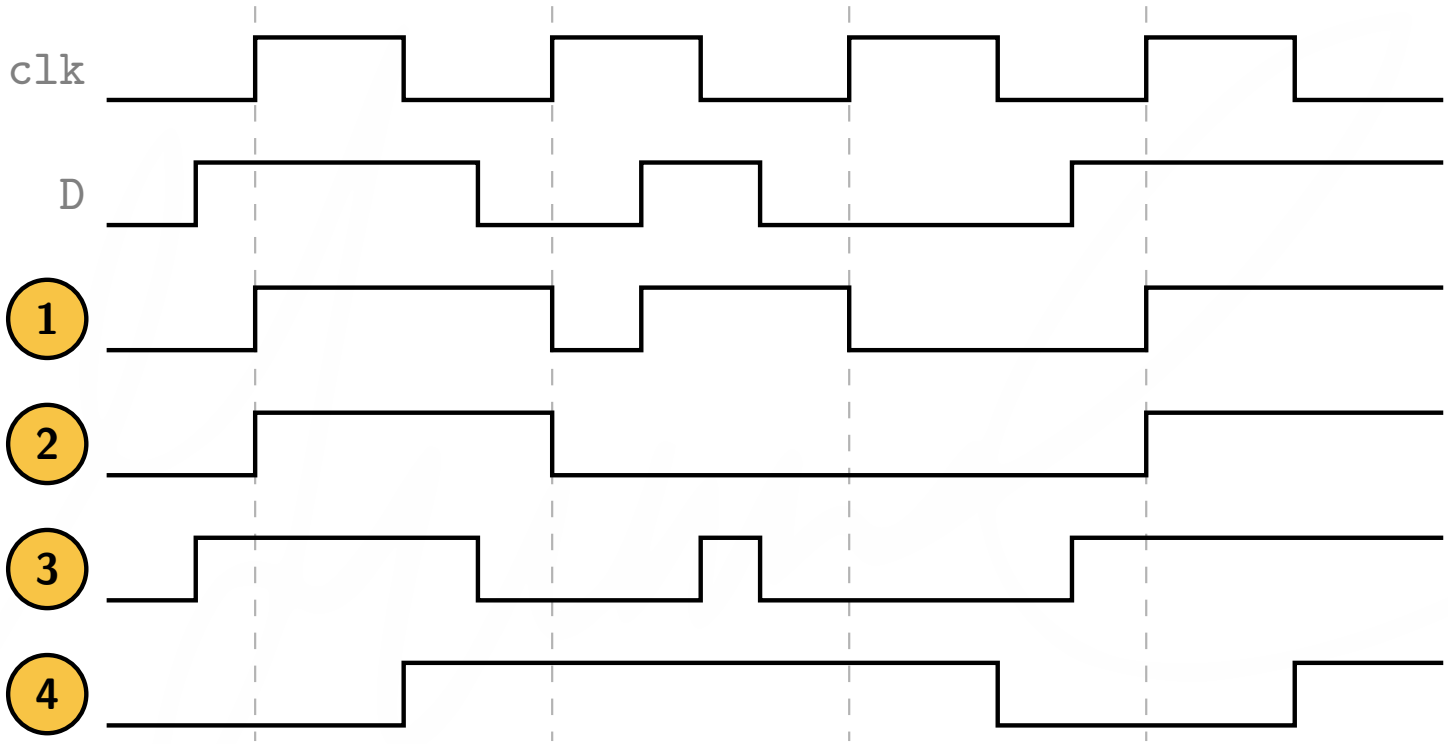
if D changes in the shaded window, the stored value is unreliable

question 13



`Q` starts at `0`; which waveform is `Q`?

question 15

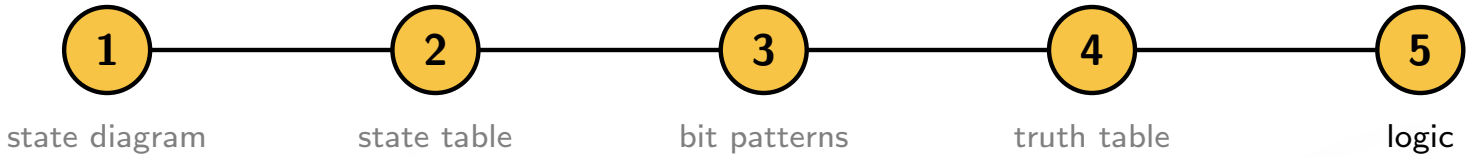


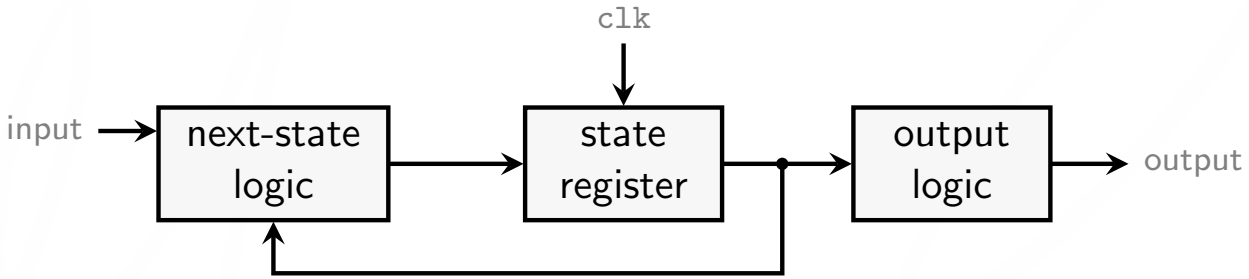
`Q` starts at `0`; which waveform is `Q`?

8

finite state machines

designing an FSM



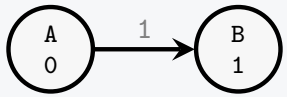


flip-flops needed

$\lceil \log_2 k \rceil$

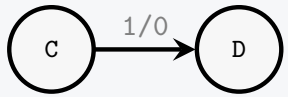
for k states (5 to 8 states need 3)

Moore



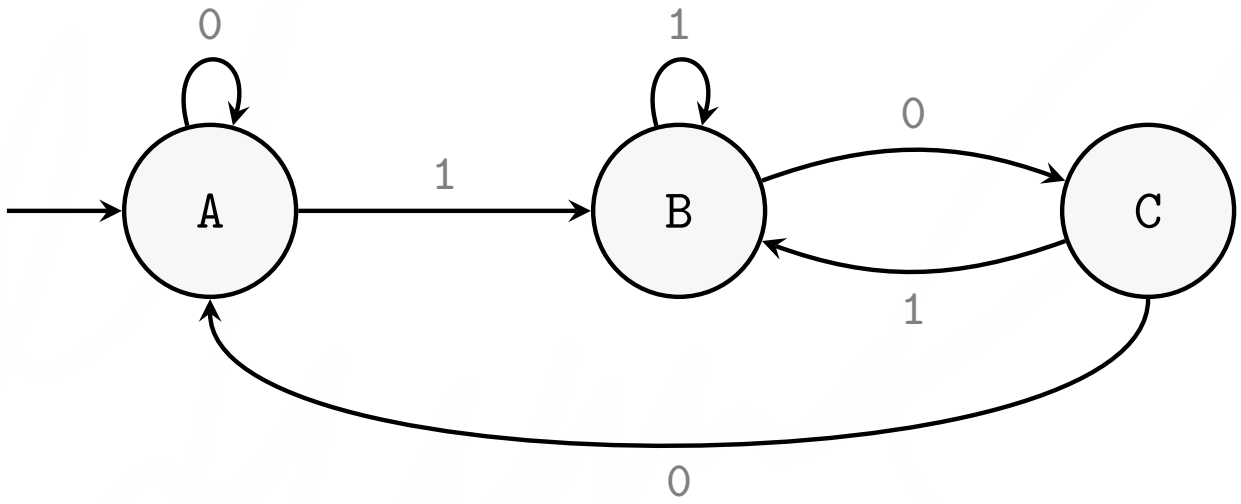
output depends on the state only

Mealy



output depends on the state and input

question 18



the machine starts in A; the input is 1, 0, 1, 1, 0, 1, 0, 0, 1

question 19

Q_1	Q_0	x	D_1	D_0
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	1	0
1	0	1	1	1
1	1	0	1	1
1	1	1	0	0

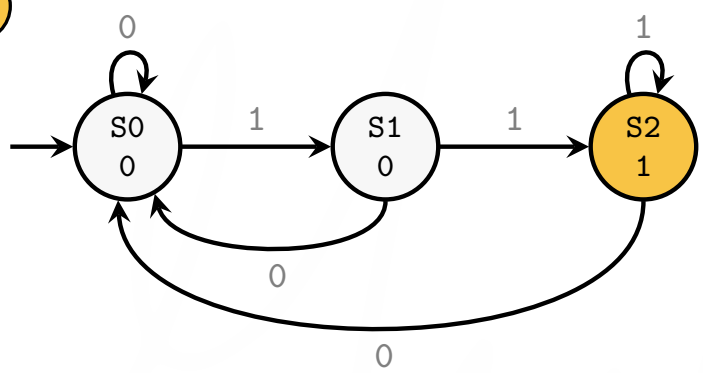
a 2-bit counter: it counts up when $x = 1$ and holds when $x = 0$

which equation gives D_1?

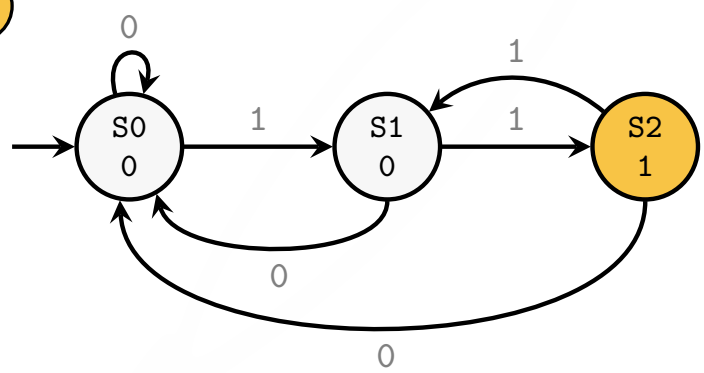
question 20

output 1 exactly when the last two inputs were both 1

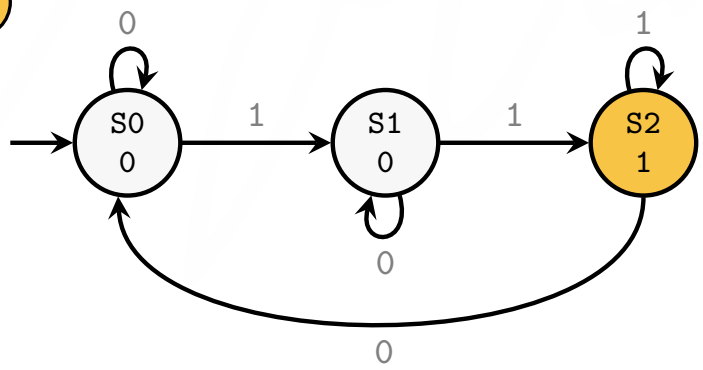
1



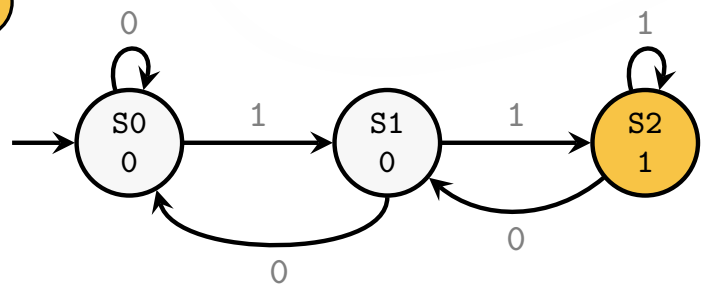
2



3



4



putting it all together

Good luck on the **midterm**!